WaveCatBoost: Hybrid Air Quality Forecasting

Applied Time Series Analysis — Course Project

Group 8

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Problem and Approach

Why this is hard

- Air-quality forecasting needs **near real-time** answers.
- That pushes us toward **lighter** models (trees, short lags).
- The series is **non-stationary**: spikes, regime shifts, and changing variance.
- Light models struggle on the **raw** mix of spikes, diurnal cycles, and slow drift.

Paper approach

1. Split the series into scale-wise components (details + smooth) with Haar MODWT.
2. Forecast each component with CatBoost.
3. Reconstruct by summing bands; optionally add conformal intervals.

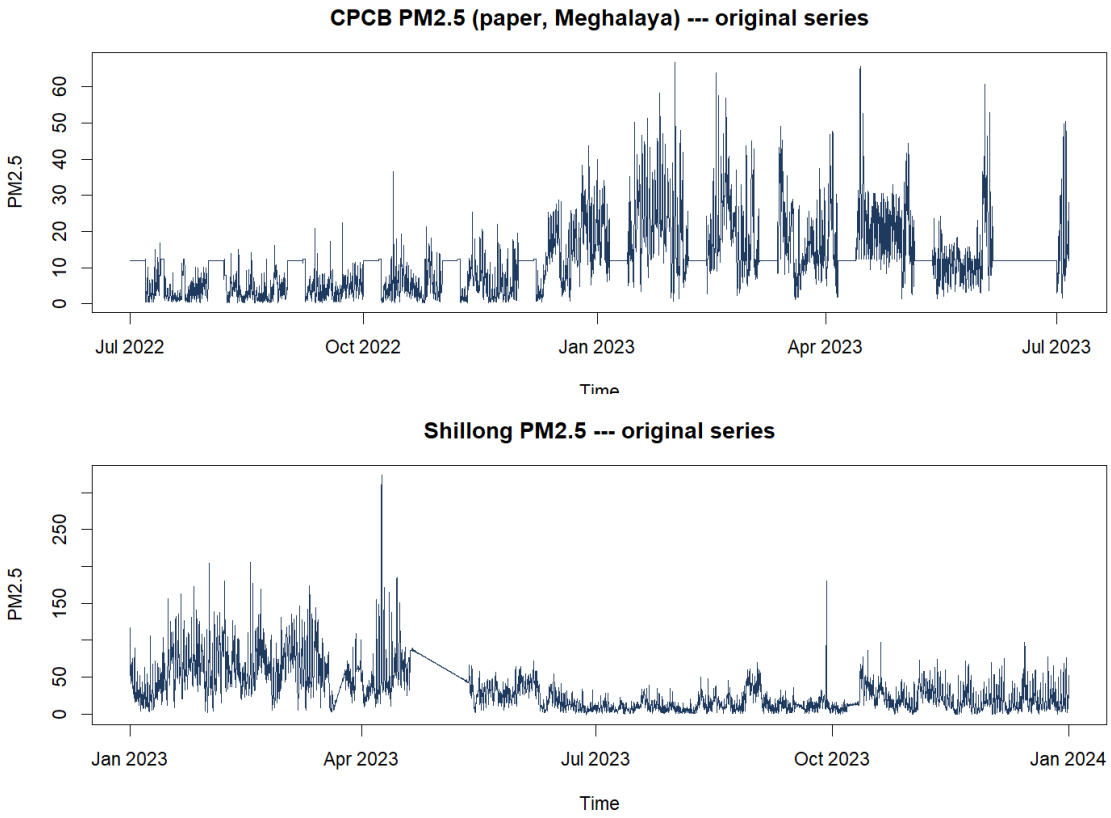
Datasets

Paper CSVs (Meghalaya, hourly)

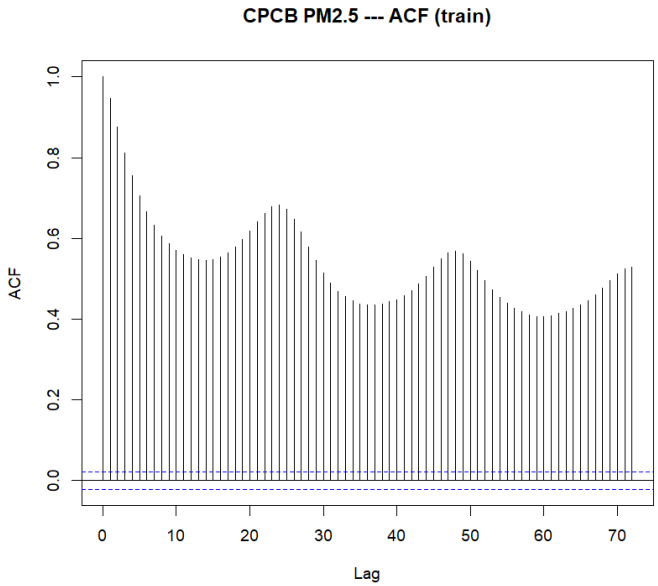
- **CPCB**, **ID1** from Borah et al.
- $n \approx 8880$ / 9505.
- NO_2, O_3, CO, SO_2, $PM_{2.5}$, PM_{10}.

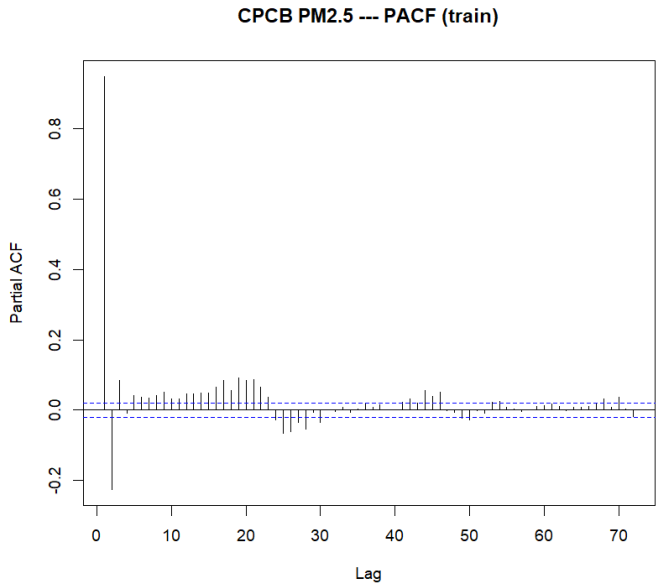
Extra CAAQMS (15-min $\rightarrow$ hourly)

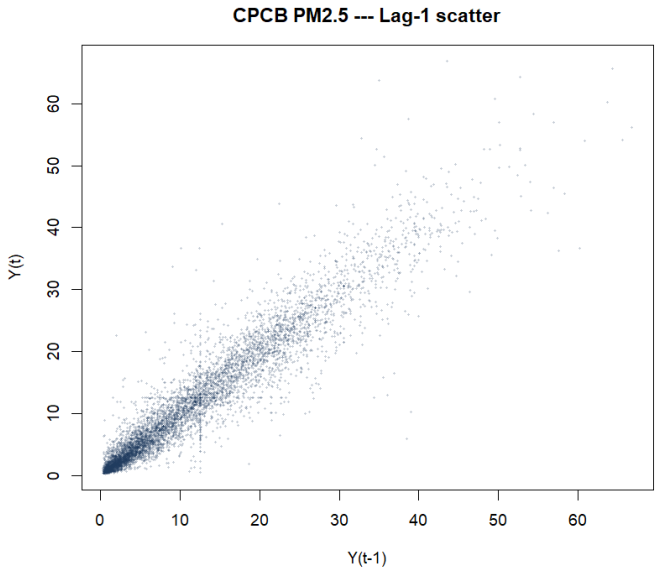
- **Shillong:** Lumpyngngad, JN Stadium.
- Delhi: ITO, R K Puram.

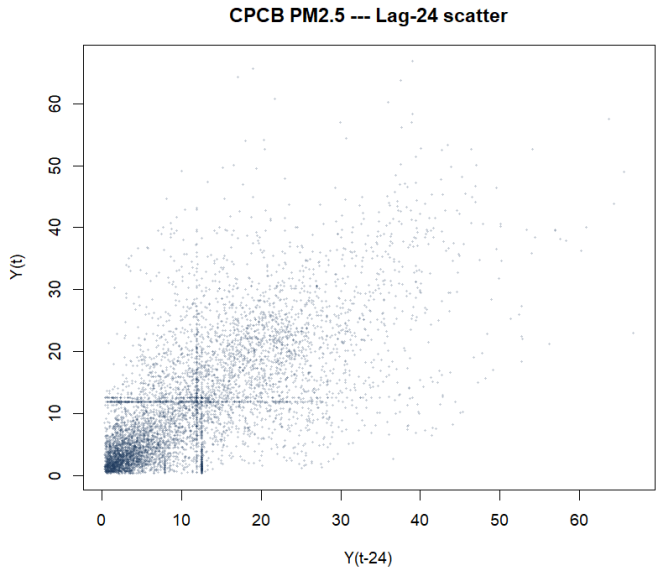


Dataset diagnostics — CPCB $PM_{2.5}$

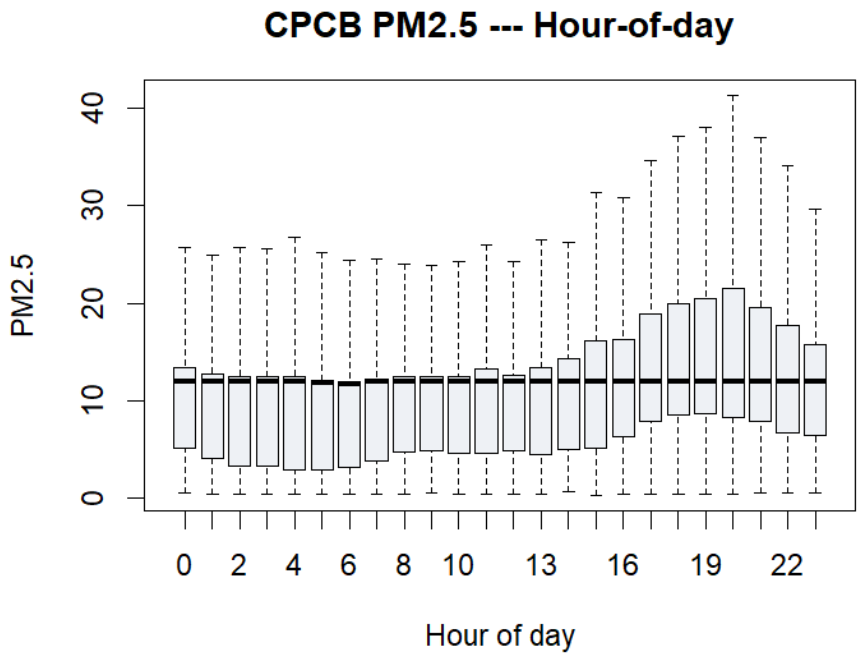


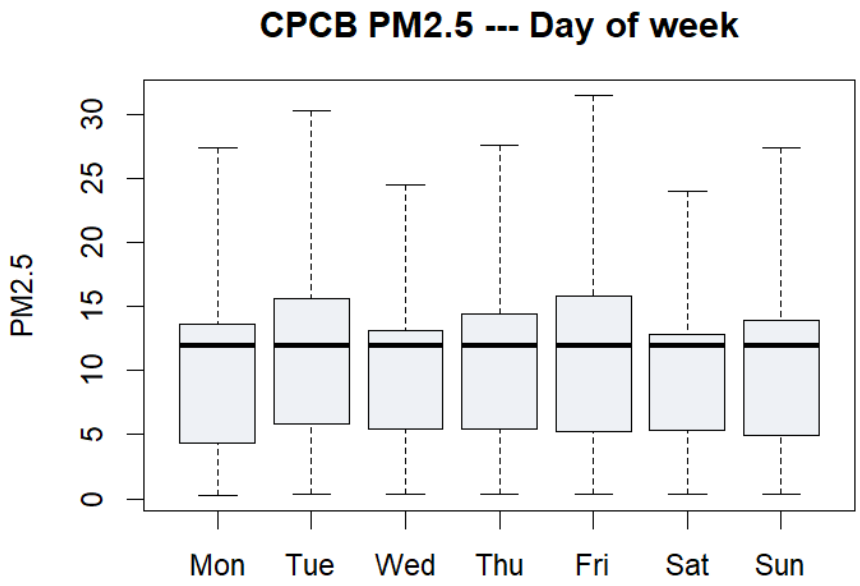






Seasonal structure — CPCB $PM_{2.5}$





- Clear diurnal pattern; milder weekday effect.
- Motivates $S = 24$ baselines and lag features up to 24 h.

What we did before modelling

- **Hourly grid:** end-labelled hourly means (15-min CAAQMS $\rightarrow$ hour t).
- **Quality screen:** physical bounds, stuck sensors, spike flags.
- **Causal fill:** impute only with information available up to the cut time.
- **Chronological split:** train $\rightarrow$ val (tuning) $\rightarrow$ test (last h hours). Horizons $h \in \{24, 168, 336, 744\}$.
- **Scale:** min–max to $[0, 1]$ fit on **train only**.
- Paper Meghalaya CSVs are already hourly $\Rightarrow$ skip re-aggregation / fill. Extra stations (Shillong, Delhi) get the full clean path into an isolated store.

Why decompose? Why not Fourier?

Why split the series

- Raw Y_t mixes noise, diurnal swings, and slow level shifts.
- One light model on the mix misses scale-specific structure.
- Decompose $\rightarrow$ forecast each band $\rightarrow$ add back $=$ richer dynamics without a heavy global model.

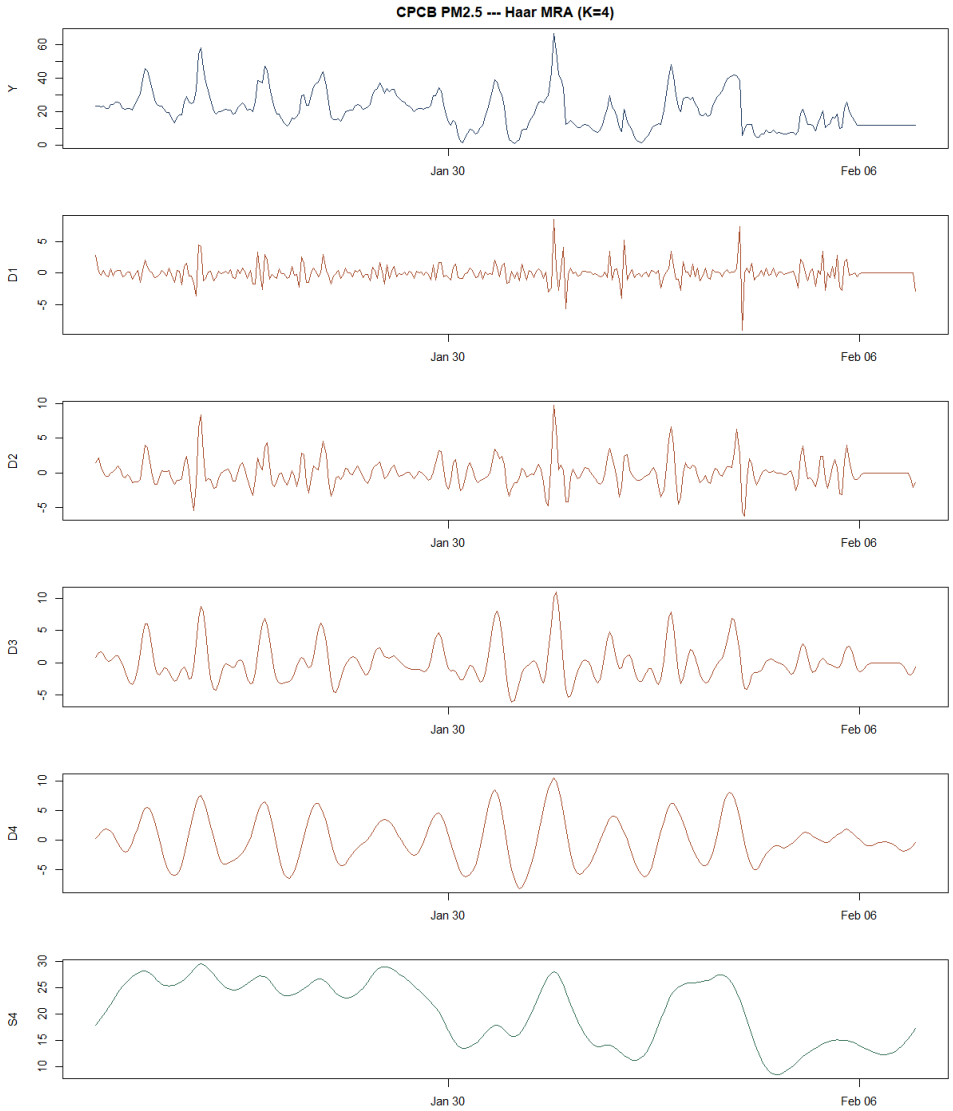
Why not Fourier

- Fourier uses **global** sinusoids over the whole record.
- Air quality is **non-stationary** — spikes and regime changes are local in time.
- Wavelets give a joint **time–scale** view; MODWT keeps every band length N (no downsampling).
- Haar MRA is additive: $Y_t = \sum_k D_{k,t} + S_{K,t}$.

MODWT: additive components

$$Y_t = \sum_{k=1}^{K} D_{k,t} + S_{K,t}$$

- $D_{k,t}$: detail at scale $\sim 2^k$ h.
- $S_{K,t}$: smooth after K levels.
- Paper: $K \approx \ln N$; we use K=4–5 ($\sim$ 16–32 h smooth).
- D_1: high-frequency component;
  D_k: lower-frequency component;
  S_K: smooth trend / level.



MODWT: mathematical intuition

Wavelet filter: $p_m \to D$ (details)
$\sum_m p_m = 0$ (measures change)

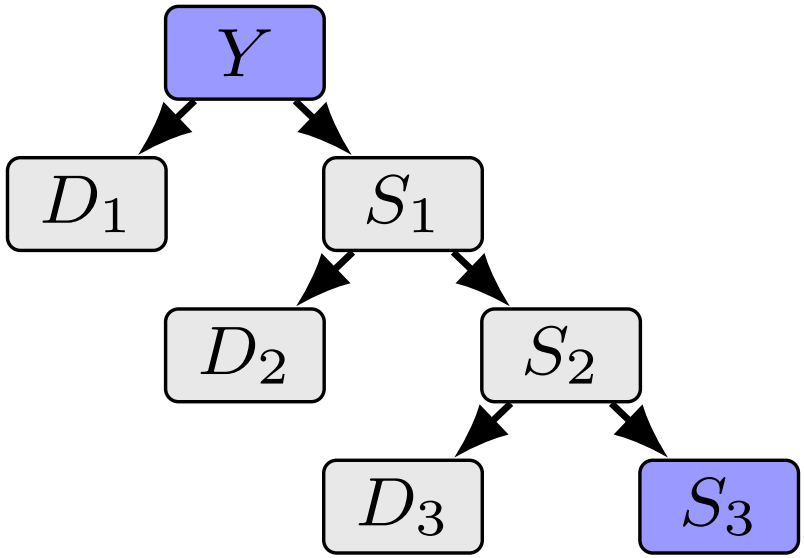
Scaling filter: $q_m \to S$ (smooth)
$\sum_m q_m = 1$ (measures average)

Filter condition (nonzero n):
$\sum_m p_m p_{m+2n} = \sum_m q_m q_{m+2n} = 0$

Recursion at level j ($S_{0,t} = Y_t$)
$D_{j,t} = \sum_m p_m \, S_{j-1,(t-2^{j-1}m) \bmod N}$
$S_{j,t} = \sum_m q_m \, S_{j-1,(t-2^{j-1}m) \bmod N}$

Haar: $q = (\frac{1}{2}, \frac{1}{2})$, $p = (\frac{1}{2}, -\frac{1}{2})$.

Tap spacing doubles each level.

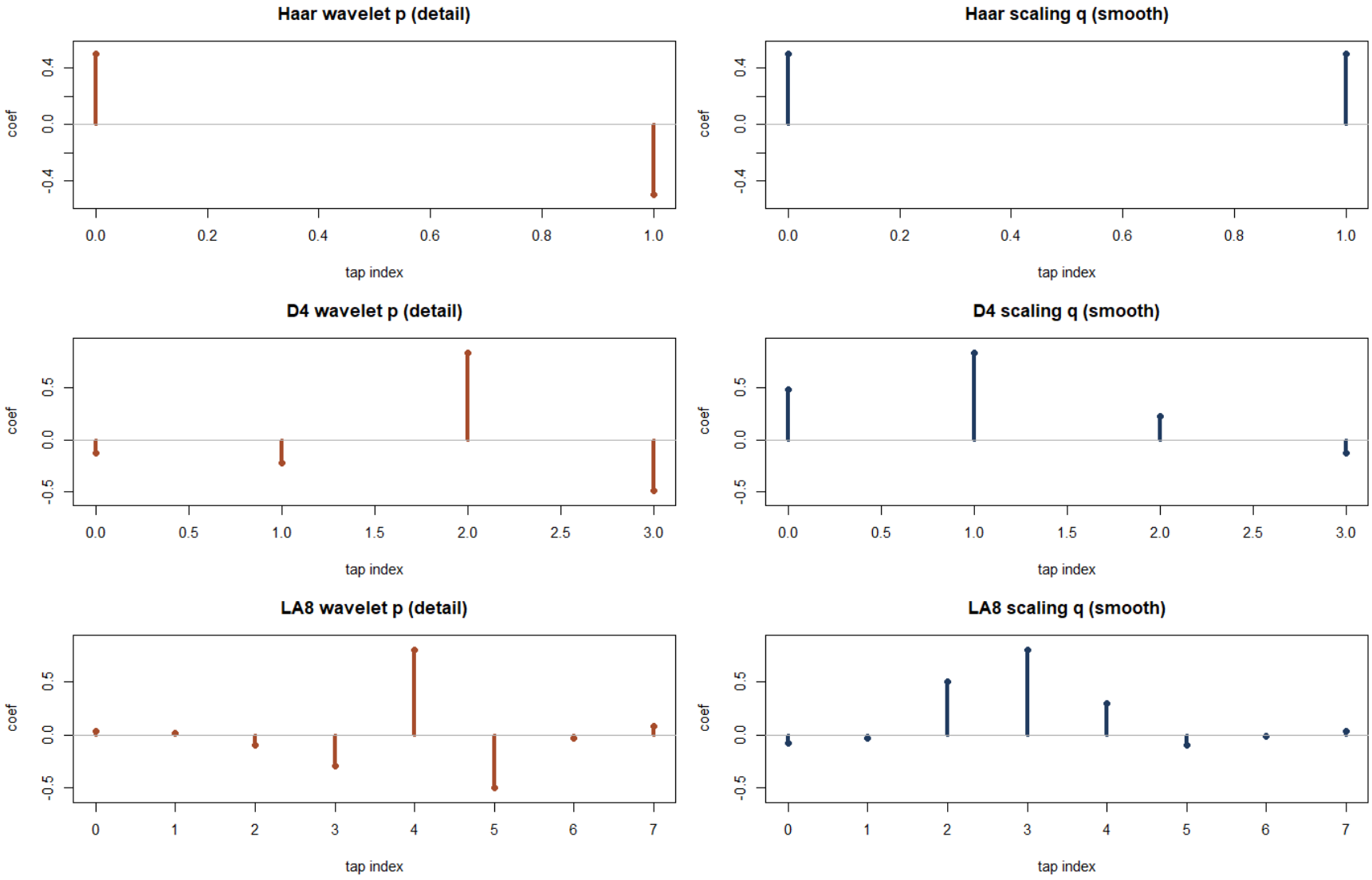


Only the smooth is split again.
$Y = D_1 + D_2 + D_3 + S_3$

Inverse (IMODWT): same filters, reversed

$S_{j-1,t} = \sum_m p_m D_{j,t+2^{j-1}m} + \sum_m q_m S_{j,t+2^{j-1}m}$

Filter coefficients



Haar vs D4 vs LA8 (WaveCatBoost, 1-day MASE)

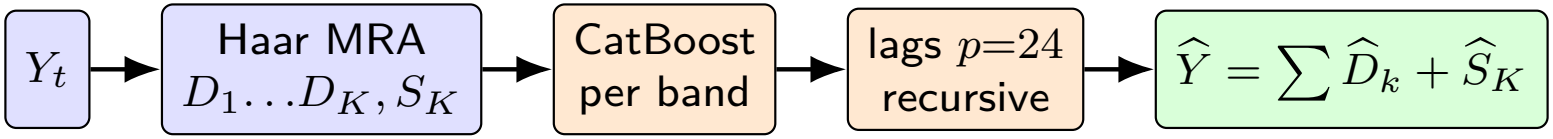
- **Paper default:** Haar ($M{=}2$).
- **We also tried:** D4 ($M{=}4$) and LA8 ($M{=}8$).
- Same WaveCatBoost head, same K and lag p.

Sensor	Pollutant	Haar	D4	LA8
CPCB	NO_2	0.541	0.510	0.529
CPCB	$PM_{2.5}$	0.727	0.909	0.955
ID1	NO_2	1.252	1.346	1.330
ID1	$PM_{2.5}$	0.421	0.457	0.502

Takeaway

Haar wins on 3 of 4 cells (D4 edges CPCB NO_2). Longer filters do not help overall — Haar stays the default.

WaveCatBoost pipeline



- One CatBoost per detail / smooth band; reconstruct by summing.
- Point forecast alone is not enough for air-quality decisions — we need uncertainty.

Why conformal prediction?

Idea in one line

WaveCatBoost gives one number $\widehat{Y}_t$ ("$PM_{2.5}$ will be 20").
Conformal adds a **band** around it: "with about 95% chance, the truth is between L_t and U_t."

- **Why we need it:** a single guess is risky for alerts — we want a safe range.
- **How it works :** look at recent forecast mistakes; if errors were large lately, make the band wider; if small, make it narrower.
- **Why not a normal-based interval?** air-quality errors are irregular (spikes, quiet stretches). Conformal does **not** assume a normal curve.
- **What "95%" means:** over many hours, about 95 out of 100 true values should fall inside the band (coverage).

Conformal formula

$$CS_t = \frac{|Y_t - \widehat{Y}_t|}{\widehat{U}_t}, \qquad CQ_t = Q_{1-\alpha}(CS_{t-\kappa+1:t}), \qquad PI_t = \widehat{Y}_t \pm CQ_t\,\widehat{U}_t$$

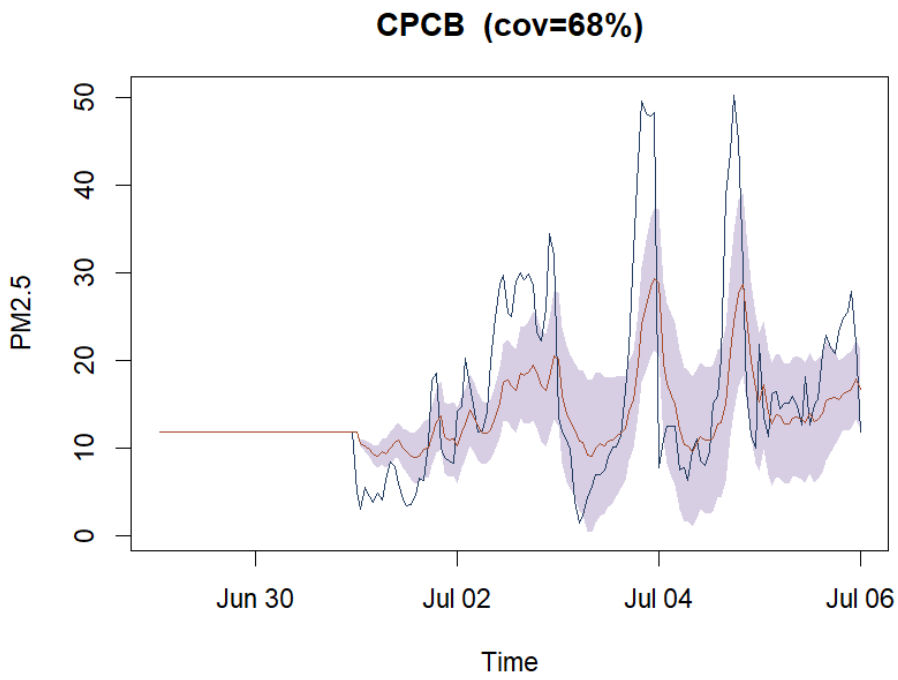
- Y_t, $\widehat{Y}_t$: observed value and WaveCatBoost forecast.
- $\widehat{U}_t$: rolling MAE of recent residuals (local error scale).
- CS_t: nonconformity — absolute error scaled by $\widehat{U}_t$.
- α: miscoverage rate ($\alpha = 0.05 \Rightarrow$ 95% target).
- κ: calibration window length (we try $\{24, 168, 336\}$ h).
- CQ_t: $(1-\alpha)$-quantile of the last κ scores CS.
- $L_t = \widehat{Y}_t - CQ_t\widehat{U}_t$, $U_t = \widehat{Y}_t + CQ_t\widehat{U}_t$ (clip $L_t \geq 0$ for concentrations).

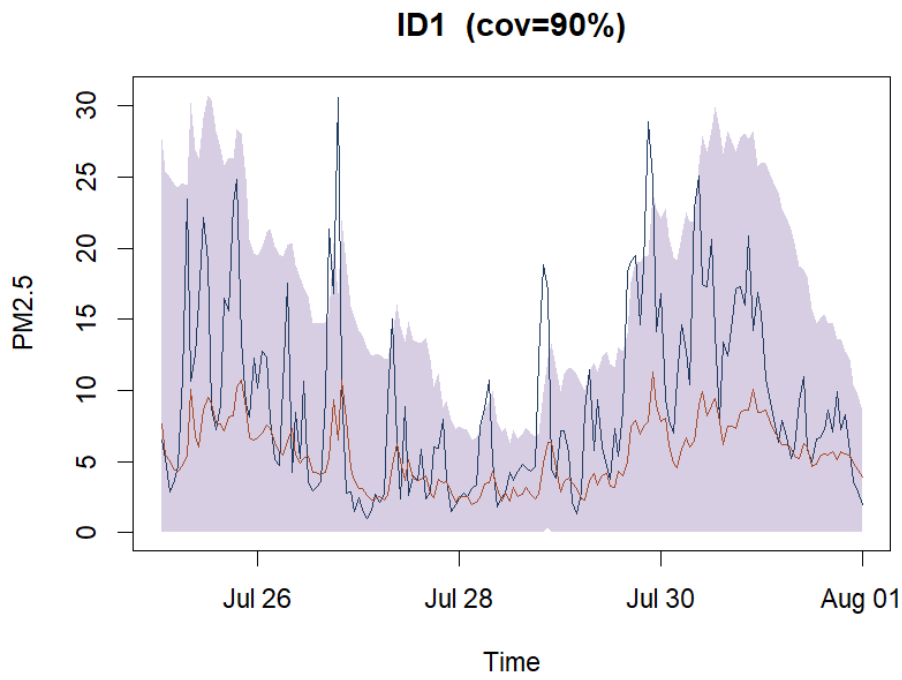
How to use it

Fit on train $\rightarrow$ compute CS on a held-out calibration window $\rightarrow$ pick κ so cal. coverage $\approx 95\% \rightarrow$ apply $CQ\,\widehat{U}$ on the test hours. Coverage = fraction of test points with $Y_t \in [L_t, U_t]$.

Do the intervals work? (7-day test)

Purpose: on held-out hours, does true $PM_{2.5}$ (blue) fall inside the purple band around the forecast (red)?





- **Blue** = truth; **red** = WaveCatBoost guess; **purple** = conformal window $[L_t, U_t]$.
- **Coverage** = % of hours where blue is inside purple (target $\approx 95\%$). ID1 $\approx 90\%$ (OK); CPCB $\approx 68\%$ (too narrow after a flat spell).

Making intervals better — CQR

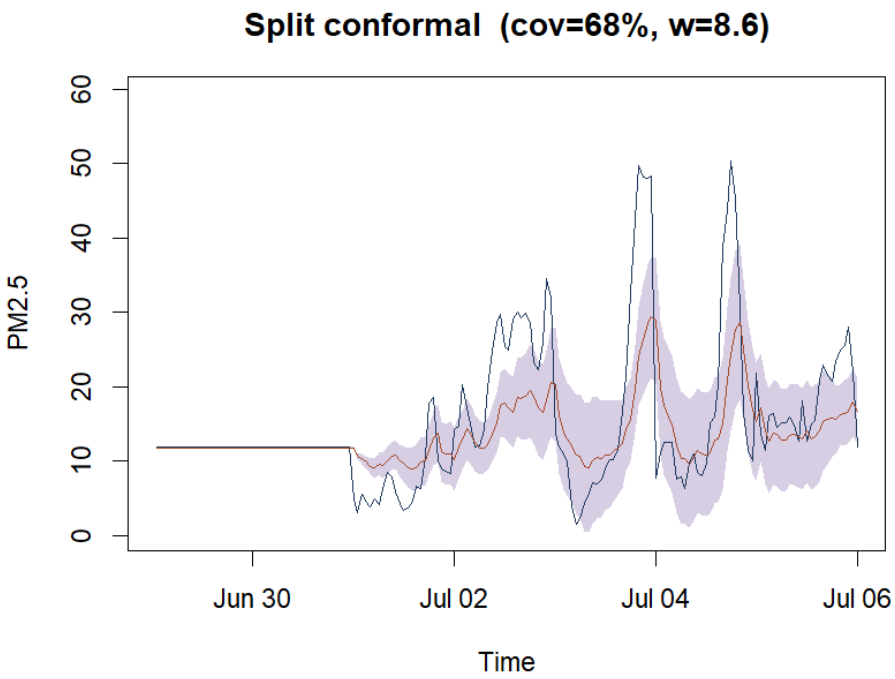
- Paper / split conformal scales one point forecast with past errors.
- **CQR** (extension): fit low & high quantile models, then conformalize $s_i = \max(\hat{q}_{lo} - Y_i,\ Y_i - \hat{q}_{hi})$.
- Same split protocol (calibrate off the test window).

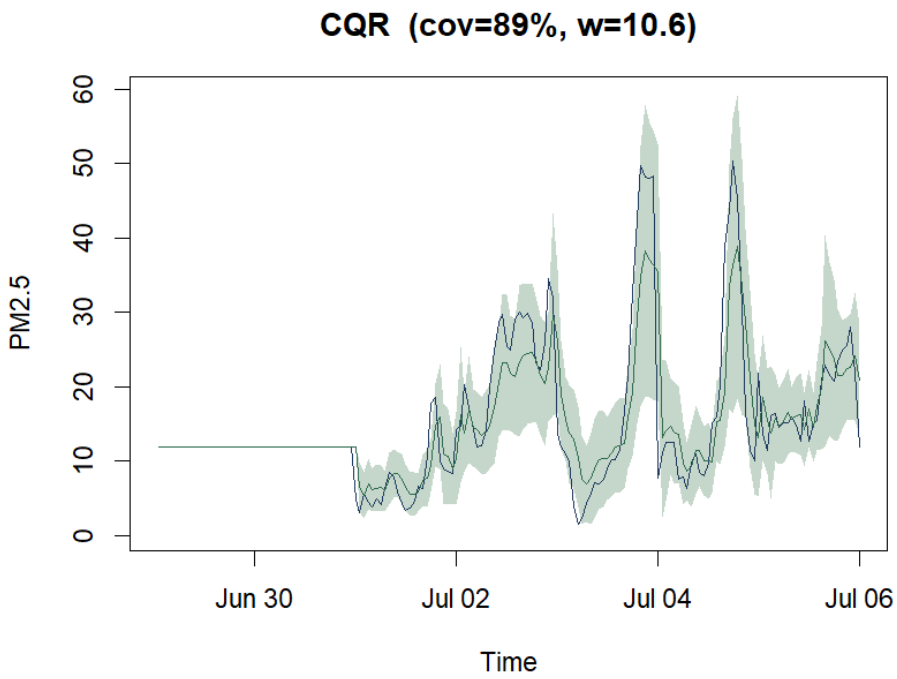
	Split conformal		CQR	
Sensor	Cov.	Width	Cov.	Width
CPCB	0.68	8.6	**0.89**	10.6
ID1	0.90	17.4	**0.98**	18.7
Shillong	0.90	127	**0.92**	**34.6**

Takeaway

CQR lifts CPCB coverage and cuts Shillong width sharply — next slide shows CPCB bands.

CQR vs split conformal (CPCB $PM_{2.5}$)





Experimental results — what we compare

Baselines vs WaveCatBoost

Same data splits and horizons; only the model changes.

- **Seasonal naive** (lag $S{=}24$), **SARIMA**
- Trees on **raw** Y: LightGBM, CatBoost (, XGBoost)
- Wave + trees: WaveLGB, **WaveCatBoost** (paper model)
- Paper also reports RNN, TCN, TFT, Transformer (Table 1)

- **Why MASE?** pollutants have different units/scales (NO_2 vs $PM_{2.5}$). MASE is **scale-free**: error relative to a seasonal naive on train, so we can compare methods across pollutants.
- Also report MAE / RMSE in original units after inverting min-max.

Error metrics (formulas)

With train length N, horizon h, seasonality $S{=}24$:

$$\mathrm{MASE} = \frac{\frac{1}{h}\sum_{t=1}^{h} |\widehat{Y}_t - Y_t|}{\frac{1}{N-S}\sum_{t=S+1}^{N} |Y_t - Y_{t-S}|}$$

(equivalent to paper Eq. (6) / Hyndman seasonal MASE).

$$\mathrm{MAE} = \frac{1}{h}\sum_{t=1}^{h} |\widehat{Y}_t - Y_t|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{h}\sum_{t=1}^{h} (\widehat{Y}_t - Y_t)^2}.$$

- MASE < 1: better than seasonal naive on the train scale.
- Lower is better for all three.

Our recursive bake-off (1-day MASE)

Sensor	Pol.	snaive	LGB	CatBoost	WaveCatBoost
CPCB	$PM_{2.5}$	2.02	0.74	0.82	**0.71**
CPCB	PM_{10}	1.46	0.52	0.75	0.53
CPCB	NO_2	0.72	0.52	0.51	0.55
ID1	$PM_{2.5}$	0.71	0.84	0.67	0.42
ID1	PM_{10}	0.66	0.81	0.73	0.41
ID1	NO_2	1.20	1.08	1.06	1.24

- **Protocol:** recursive multi-step from one origin (`scores.csv`).
- Fair for "does wavelet help vs raw CatBoost / LGB?" — **not** Table 1.

Paper Table 1 (letter)

Pollutant	Horizon	CPCB Sensor								ID1 Sensor							
		RNN	LGBM	TCN	TFT	Transformer	XGB	CatBoost	WaveCatBoost	RNN	LGBM	TCN	TFT	Transformer	XGB	CatBoost	WaveCatBoost
NO_2	1d	2.45	0.79	1.45	1.41	3.28	0.91	0.82	**0.57**	2.05	1.27	2.15	1.62	2.07	1.40	1.20	**0.91**
	7d	1.35	**0.66**	0.67	2.65	0.93	**0.66**	1.04	0.84	1.66	0.82	1.64	1.76	1.76	0.95	0.84	**0.52**
	14d	0.81	**0.71**	1.79	4.27	2.05	0.72	0.73	0.72	1.57	0.85	1.56	2.06	1.63	0.92	1.69	**0.42**
	31d	1.98	0.75	1.87	1.18	1.35	0.76	1.17	**0.47**	1.47	1.15	1.48	1.95	1.86	1.14	1.36	**0.40**
O_3	1d	5.19	1.87	1.85	2.43	11.47	1.52	2.69	**0.98**	3.65	1.61	3.32	2.79	3.75	1.72	1.49	**1.42**
	7d	5.46	1.68	4.66	6.16	5.82	1.51	4.84	**0.76**	3.57	1.64	2.55	2.84	3.18	1.96	1.79	**0.72**
	14d	6.24	1.23	4.38	14.68	9.54	1.27	1.77	**1.11**	3.85	1.80	3.67	3.74	3.12	2.06	2.32	**0.64**
	31d	3.56	2.46	4.26	2.71	3.56	2.44	5.37	**0.25**	3.59	2.72	3.63	3.56	3.61	2.87	2.66	**1.11**
CO	1d	2.36	3.36	2.90	2.45	**0.97**	3.21	2.87	2.69	4.47	2.48	3.82	2.95	4.47	2.54	2.32	**1.53**
	7d	3.74	3.62	3.94	5.29	3.74	3.59	3.57	**2.61**	2.85	1.52	2.64	2.95	2.84	1.46	1.36	**0.79**
	14d	2.76	3.06	2.79	4.34	2.77	2.88	3.15	**1.63**	3.08	1.93	3.08	3.71	3.06	1.83	3.36	**0.83**
	31d	1.92	2.05	1.96	3.29	1.91	1.98	2.03	**0.61**	2.79	2.58	2.79	3.48	3.57	2.22	3.26	**0.71**
SO_2	1d	34.35	11.46	28.11	**7.76**	40.15	10.62	9.35	23.93	1.81	0.45	0.85	0.49	1.79	**0.43**	0.44	1.31
	7d	16.83	15.43	14.62	20.72	11.59	14.39	**5.84**	7.30	2.36	0.34	2.23	0.61	2.21	0.33	**0.30**	0.61
	14d	10.70	7.35	9.37	24.75	9.57	8.14	11.31	**6.17**	2.15	0.22	2.82	1.66	3.18	0.26	**0.24**	0.58
	31d	9.06	11.22	10.34	11.44	10.79	9.13	9.81	**5.33**	2.35	0.41	2.95	0.81	1.88	**0.26**	0.31	1.31
$PM_{2.5}$	1d	2.88	3.81	**2.50**	3.52	6.57	2.95	3.02	5.06	7.31	4.67	7.03	4.68	7.35	5.53	4.45	**0.85**
	7d	6.94	6.86	7.05	7.38	6.69	5.92	6.57	**5.07**	7.09	3.01	5.22	4.22	5.56	2.65	2.43	**0.58**
	14d	4.40	6.98	4.40	7.07	4.41	6.20	6.30	**2.34**	8.20	4.43	8.71	7.10	7.39	3.67	5.58	**0.69**
	31d	2.73	2.85	2.83	4.41	2.73	2.83	2.83	**1.57**	7.21	5.26	7.01	5.97	5.05	4.49	5.29	**0.42**
PM_{10}	1d	**1.01**	6.69	1.79	6.37	4.16	3.81	5.77	3.12	7.31	4.34	7.06	4.24	7.35	5.59	4.22	**0.86**
	7d	5.20	5.30	5.32	6.19	5.11	4.99	5.24	**3.70**	7.37	2.83	5.42	4.34	5.81	2.46	2.27	**0.57**
	14d	3.53	4.59	3.55	5.90	3.54	4.36	4.89	**1.48**	8.48	4.29	9.08	7.08	7.65	3.29	5.44	**0.67**
	31d	2.39	2.40	2.51	3.64	2.37	2.43	2.40	**0.98**	7.57	5.39	7.35	6.09	5.36	4.50	5.47	**0.41**

time-index are computed by considering previous time steps in the σ_r permutation, i.e., $\text{grad}_{r,\sigma(i)-1}(i)$, where

$$\text{grad}_{r,j}(i) = \left.\frac{\partial L\left(y_i, A\right)}{\partial A}\right|_{A=M_{r,j}(i)}. \tag{3}$$

The leaf value for the ith instance, used for evaluating candidate splits, is obtained by averaging the corresponding gradients grad of

III. EXPERIMENTAL RESULTS

This section presents the experimental results showcasing the efficacy of our proposed WaveCatBoost framework compared to benchmark forecasting methods. In our analysis, we compare the performance of the proposed WaveCatBoost architecture with state-of-the-art forecasting methodologies including light gradient-boosting machines (LGBM), extreme gradient-boosting (XGB), Transformer,

Letter: rolling 1-step MASE. WaveCatBoost vs RNN / LGBM / TCN / TFT / Transformer / XGB / CatBoost.

1-step re-run vs Table 1 (full)

Letter-style protocol: full-series Haar MODWT + walk-forward **1-step** (`run_paper_protocol_1d.R`).

		Paper Table 1			Our 1-step re-run		
Sensor	Pol.	LGB	CatB	WaveCB	LGB	CatB	WaveCB
CPCB	$PM_{2.5}$	3.81	5.32	0.57	0.69	0.63	**0.40**
CPCB	NO_2	0.79	0.82	0.57	0.57	0.52	**0.39**
CPCB	O_3	1.87	2.89	0.98	0.33	0.33	**0.17**
ID1	$PM_{2.5}$	7.31	4.45	0.85	0.19	0.20	**0.09**

- Our WaveCB MASE is **lower** than Table 1 on every cell.
- Our LGB / CatBoost are also much lower — the whole scale shifted.

Did we beat the paper?

What we can say

- **Ranking matches Table 1:** WaveCatBoost beats LGB and CatBoost on every overlapping 1-step cell.
- Our WaveCatBoost MASE is numerically smaller than the letter's WaveCB.

What we should *not* claim

"We beat the paper" as a scientific result — baselines dropped too (e.g. CPCB $PM_{2.5}$ LGB: paper 3.81 $\rightarrow$ ours 0.69). Absolute levels are **not digit-comparable** (hyperparams, notebook details, windowing).

Talk line

Under the letter's rolling 1-step setup we reproduce the same ranking — WaveCatBoost wins — and our WaveCB MASE is lower than Table 1, but we compare **gaps / ranks**, not absolute digits.

Our 1-step results (all models)

Sensor	Pol.	snaive	LGB	CatBoost	WaveCatBoost
CPCB	$PM_{2.5}$	2.02	0.69	0.63	**0.40**
CPCB	NO_2	0.72	0.57	0.52	**0.39**
CPCB	O_3	1.22	0.33	0.33	**0.17**
ID1	$PM_{2.5}$	0.71	0.19	0.20	**0.09**

- Fair within-protocol bake-off: WaveCatBoost is clearly best.
- Same qualitative conclusion as the letter.

Summary

- WaveCatBoost = Haar MODWT + CatBoost per band + reconstruct.
- Haar remains a strong default vs D4/LA8.
- Conformal bands add uncertainty; CQR improves coverage / width where split conformal struggles.
- **Recursive bake-off:** wavelet helps vs raw trees on several cells.
- **1-step vs Table 1:** same ranking (WaveCB wins); our MASE is lower, but absolute digits are not a fair "we beat the paper" claim — compare ranks/gaps.

References I



J. Borah *et al.*, "WaveCatBoost for probabilistic forecasting of multivariate time series data," *IEEE Sensors Letters*, 2025.
Public code: WaveCatBoost-Time-Series-Model.



L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," *NeurIPS*, 2018.